

SFWR ENG 3A04: Software Design III - Large System Design

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Course Outline –Highlights–

➡ Outline

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**SFWR ENG 3A04:
Software Design III
- Large System
Design**

Dr. R. Khedri

Outline

Course Objectives
and Content

Textbooks

Project, and Exams

- 1 Course Objectives and Content
 - Knowledge and Skills
- 2 Textbooks
- 3 Project, and Exams

Course Outline –Highlights–

➡ Course Objectives and Content

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Calendar Description

Sustainable architectures; design for change and expansion; software architecture design space; object oriented analysis and design; architectural styles; methodology of making architecture decisions; project organisation.

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Course Learning Objectives

- The course provides a **comprehensive** coverage of **software architectural styles**.
- It provides students with a **foundation** that can be used **to design large and sustainable** software systems that exhibit high dependability and security qualities.

Course Outline –Highlights–

➡ Course Objectives and Content

➡ Knowledge and Skills

Upon completion of this course, participants should **know and understand**

- Software requirements and its role in determining the architecture design
- Software Architecture Design Space (Types of Software Structures, Software Elements, Software Connectors)
- Design Principles and their role in leading to sustainable and dependable system

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The main architectural styles



(a) Art Deco Architecture



(b) Baroque Architecture



(c) Gothic Architecture



(d) Islamic Architecture

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The main architectural styles –Continued–



(a)
Mediterranean
Revival
Architecture



(b) Modern
Architecture



(c) Romanesque
Architecture



(d) Victorian
Architecture

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The main architectural styles –Continued–

- The main **SOFTWARE Architectural** styles
 - Batch Sequential,
 - Pipe & Filter,
 - Process-Control,
 - Repository,
 - Blackboard,
 - Main/Subroutine,
 - Master/Slaves,
 - Layered Architecture,
 - Virtual Machine,
 - Model-View-Controller,
 - Presentation-Abstraction-Control (PAC),
 - Client/Server,
 - Multi-tier,
 - Service-Oriented Architecture,
- Heterogeneous Architecture (Methodology of Architecture Decision, Quality Attributes, Selection of Architecture Styles, Evaluation of Architecture Designs)

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Upon completion of this course, participants **should be able to**

- Select among system development documentation templates
- Select among design methodologies
- Assess the impact of a requirement on the architecture design
- Identify any discrepancies in applying design principles in a proposed design
- Identify the *applicable domain*, *benefits*, and *limitations* of each one of the architectural styles (??)

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Should be able to –Continued–

- Articulate the detailed design of the main components of each one of the architectural styles
- Enhance the sustainability of the designed system
- Trace a proposed architecture to the requirements
- Properly document a system from its requirement to its detailed design with emphasis on documenting the architecture

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➡ Knowledge and Skills

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Should be able to –Continued–

- Use Computer Aided Software Engineering (CASE) tools to carry through a term software project
- Organise and plan the development of a software system
- Evaluate design solutions with regard to a specific set of requirements
- State the design principles

Course Outline –Highlights–

➡ Textbooks

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- The course is quite practical in flavour and it does point to the use of the introduced concepts in software systems design. A certain degree of previous programming, design, and mathematical experience is essential for survival.

- **Textbooks:** Kai Qian, Xiang Fu, Lixin Tao, Chong-wei Xu, and Jorge Diaz-Herrera. *Software Architecture and Design Illuminated*, Jones and Bartlett Publishers, ISBN: 9780763754204.
- **Avenue to Learn:** The latest version of the course outline, the slides used in class, the material related to the course project or to the tutorial sessions, and announcements at the course website on **Avenue to Learn** (<https://avenue.cllmcmaster.ca>)

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● Project

- A team must submit a unique deliverable per due date
- Within a team, the grade assigned to a member is proportional to her/his contribution
- The instructor reserves the right to re-evaluate the contribution of a student via an oral examination
- There will be four (4) deliverables; Each is worth 10%
- More regulations and deadlines are in the course outline